

KEVIN CHAN

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EDUCATION

Nanyang Technological University, Singapore (NTU) Aug 2024 – May 2028
Bachelor of Science with Honours (Distinction) in Mathematical and Computer Sciences (Double Major)

EXPERIENCE

Data Science (NLP) Intern | Ministry of Defence of Singapore May 2026 – Jul 2026

- Built a 9-stage NLP pipeline distilling 13.6M Reddit posts into 737K semantic chunks across 550K National Service documents, hierarchically clustered with BERTopic (UMAP + HDBSCAN) into a 359 topic taxonomy.
- Fine-tuned sentiment and 2-step commitment transformer classifiers with targeted annotation, leakage-free splits, lifting accuracy from 66% model baseline to 84% (macro F1 0.78) on a human gold testset.
- Developed a RAG chatbot over a 737K-vector FAISS index with reranked retrieval and LLM synthesis, routing queries to a pre-aggregated fact table in <100ms, deployed on a seven-page Streamlit dashboard.

PROJECTS

AI Sentiment Interview Caller [\[GitHub Link\]](#) Jun 2026 – Jul 2026

- Engineered a real-time multimodal AI voice agent (Pipecat, FastAPI), conducting live public interviews in 4 languages over Twilio telephony and browser WebRTC, per-utterance language detection and voice swapping.
- Optimised audio response delay from ~3.4s to 250ms (~0.5s per turn latency) by streaming TTS over WebSockets and running GPU-accelerated Whisper-Singlish STT and MERaLiON-SER emotion models locally.
- Implemented consent-first adaptive interview, SileroVAD barge-in, automated NRIC/phone PII redaction with a live SSE interviewer dashboard, CSV/Excel/JSON export and end-to-end Twilio wire-protocol test suite.

Joga Bonito - World Cup 2026 AI Prediction Engine [\[Live Site\]](#) [\[GitHub Link\]](#) Apr 2026 – Jun 2026

- Trained a gradient-boosting match outcome classifier and Poisson-based goal model on a custom time decayed ELO rating engine computed over 49,000+ matches, generating 10,000 Monte Carlo projections.
- Engineered a multi-pass fuzzy entity resolution pipeline to merge fragmented datasets into a 17,000+ player database, powering 18 feature layers fused with a K-Means tactical clustering for positional archetypes.
- Productionised the full stack on Vercel (Next.js) + Render (FastAPI) on 2256 matchup predictions, live 48-team bracket simulations, and mini games, shipped as a low-latency product used by real public visitors.

Context Aware Risk Engine (CARE) [\[GitHub Link\]](#) Feb 2026 – Mar 2026

- Built a Bayesian Value-at-Risk engine running 10,000 Monte Carlo simulations to set dynamic traffic safety risk thresholds, with a persistence gate requiring consecutive breaches to suppress false positives.
- Scored unstructured road imagery with zero-shot CLIP (PyTorch) via engineered semantic anchors and out-of-domain suppression, fused into a Streamlit telemetry dashboard, validated on scenario testbenches.

ACHIEVEMENTS & LEADERSHIP

DSTA BrainHack 2026 TIL-AI Advanced Track | Finalist May 2026 – Jun 2026

- Developed ML models across 4 tracks. Fine-tuned Qwen3 (multilingual speech recognition), trained YOLOv8 (object detection), TF-IDF RAG pipeline for document QA, with TensorRT/ONNX/FP16 inference optimisation.
- Engineered a heuristics-based autonomous game agent over 16+ iterations, improving competition scores from 0.07 to 0.82+, served via a six-container Docker Compose stack under a unified async API.

Orientation Aetheris | Honorary General Secretary Aug 2025 – Present

- Co-organised and led a large-scale university orientation for 200+ incoming freshmen.

LANGUAGES & TECHNOLOGIES

- Languages: Python, Java, SQL, TypeScript, R
- ML & Data: PyTorch, HuggingFace Transformers, Scikit-learn, Pandas, NumPy, FAISS, OpenCV, PowerBI